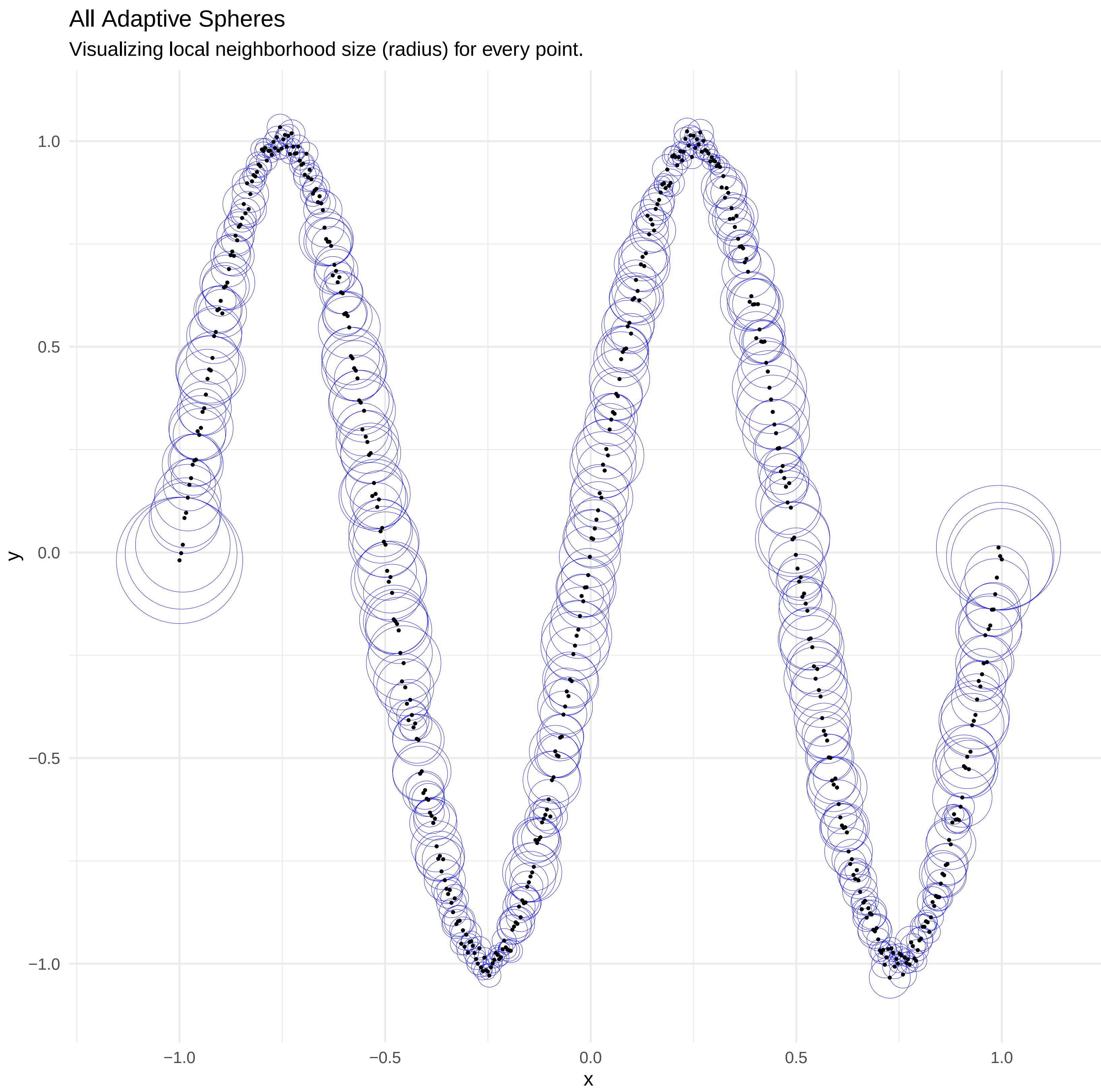
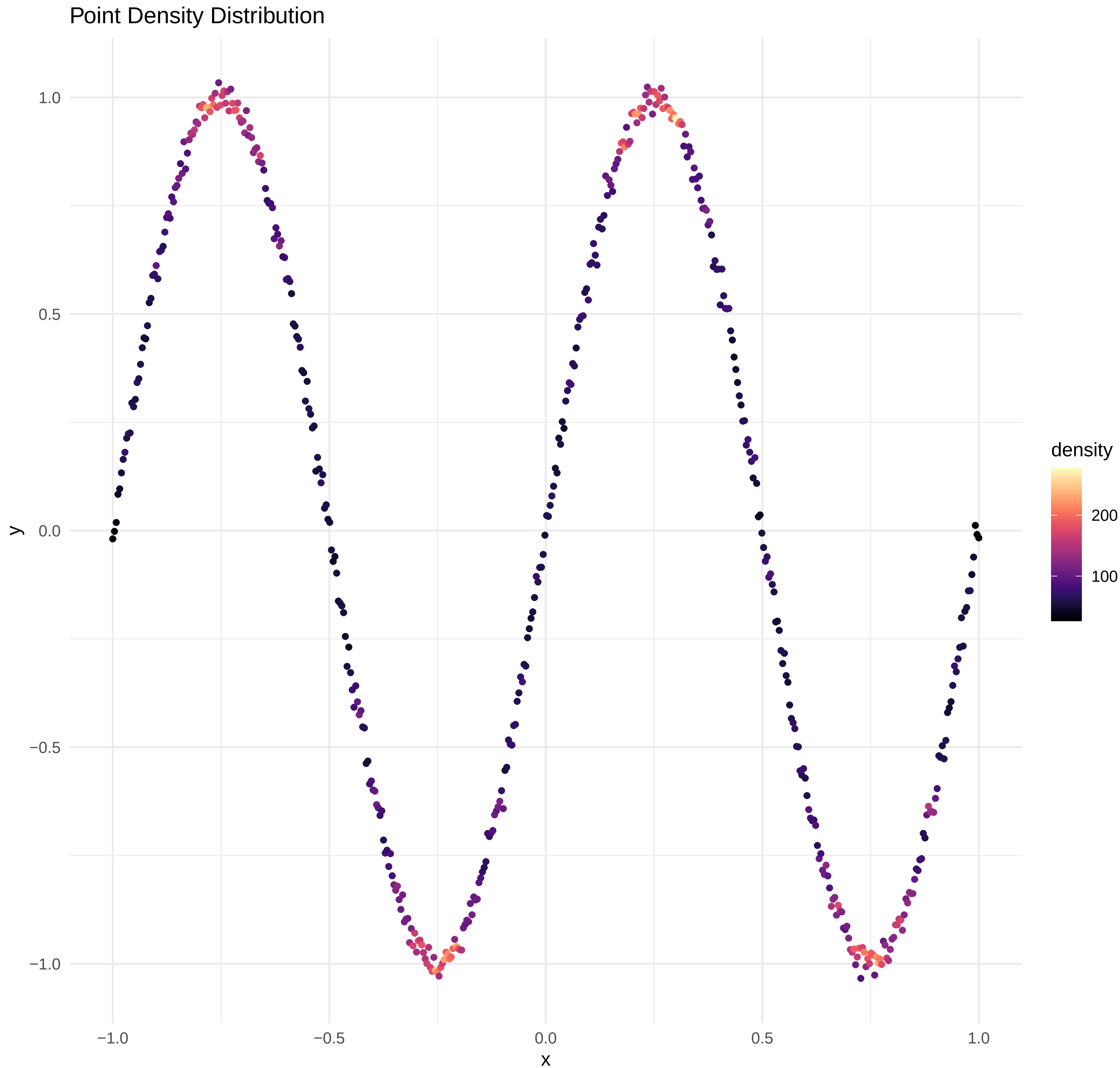
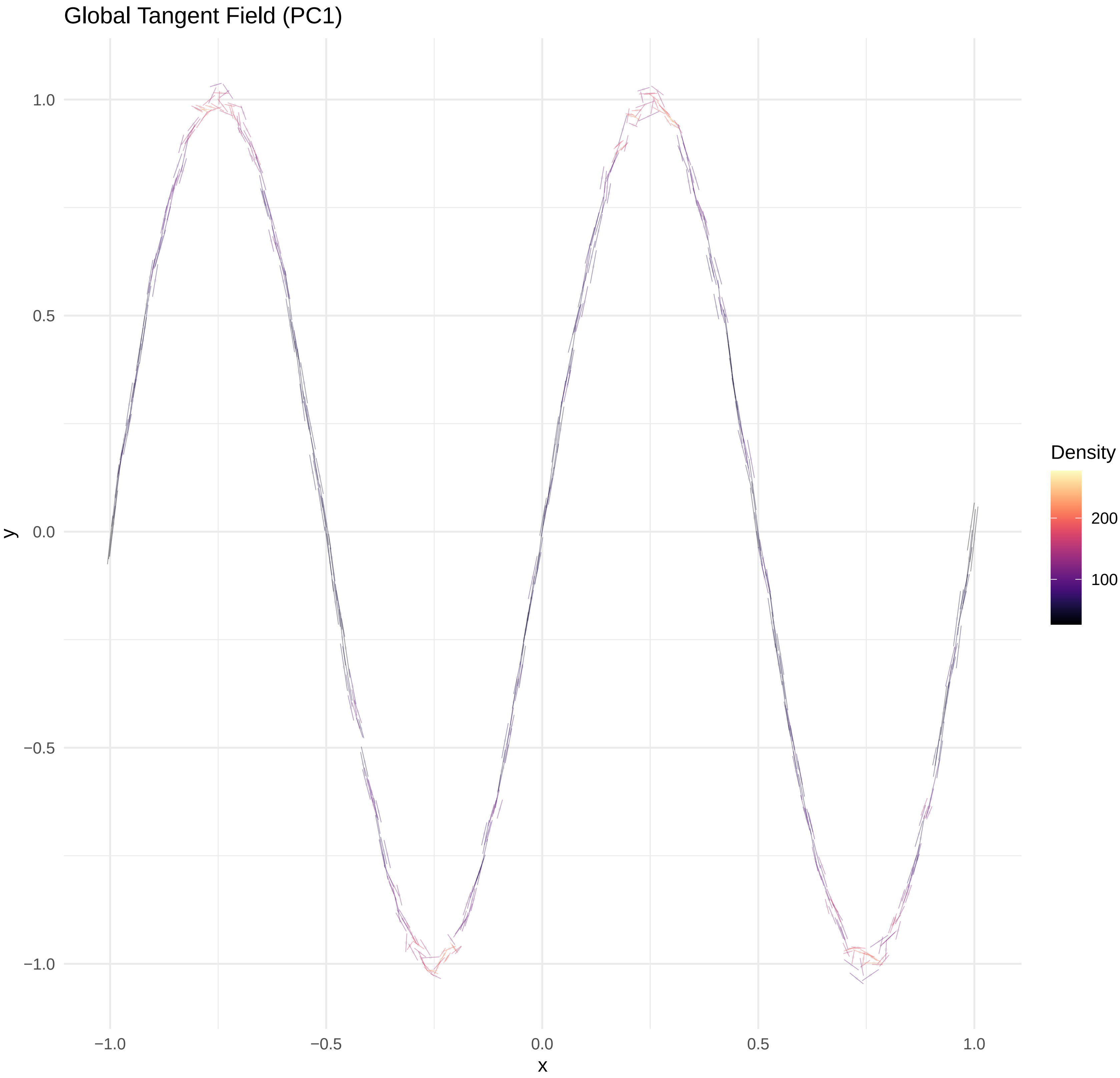
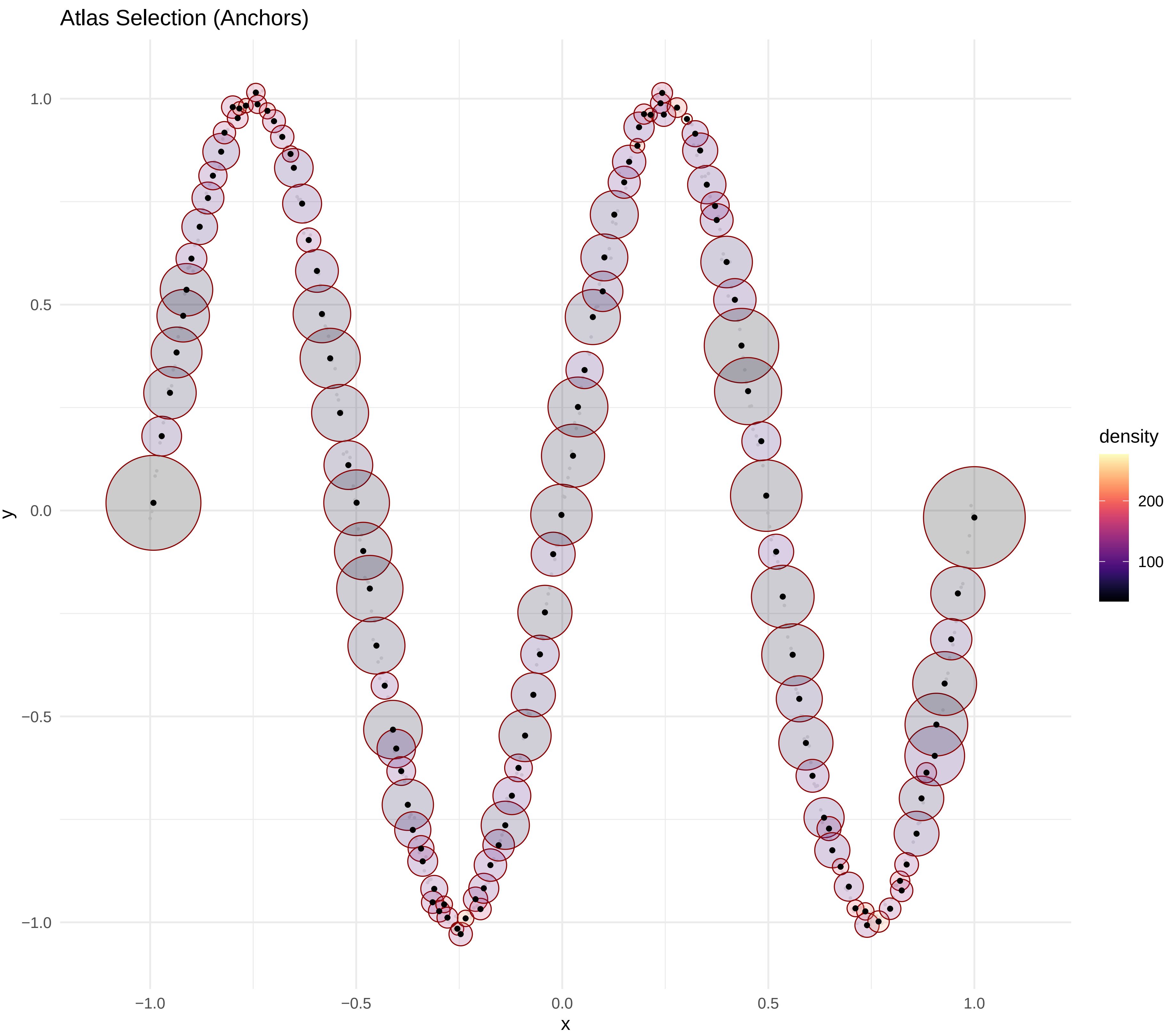


All Adaptive Spheres
Visualizing local neighborhood size (radius) for every point.
1.0
0.5
0.0
−0.5
−1.0
y
−1.0
−0.5
0.0
0.5
1.0
x

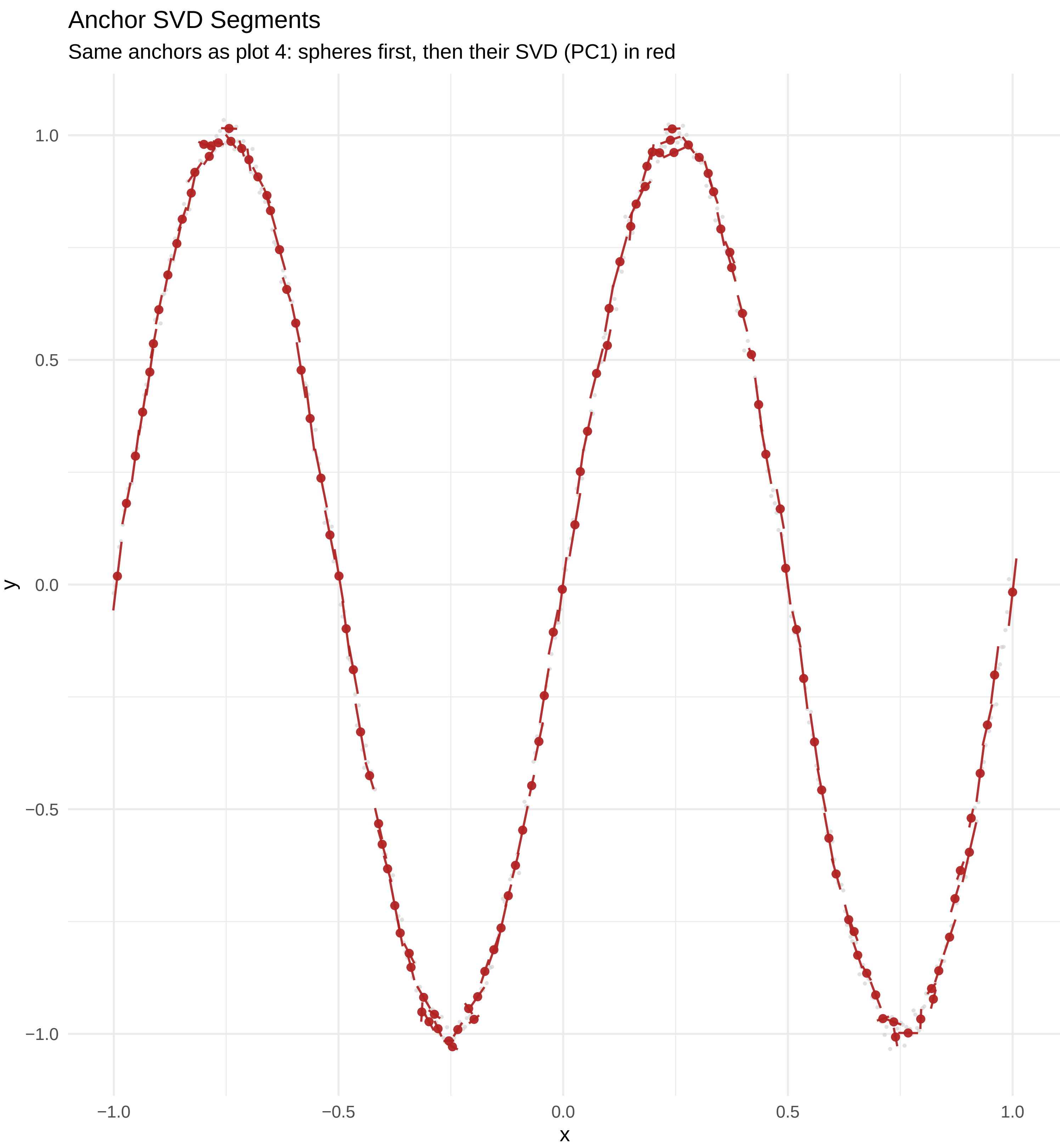


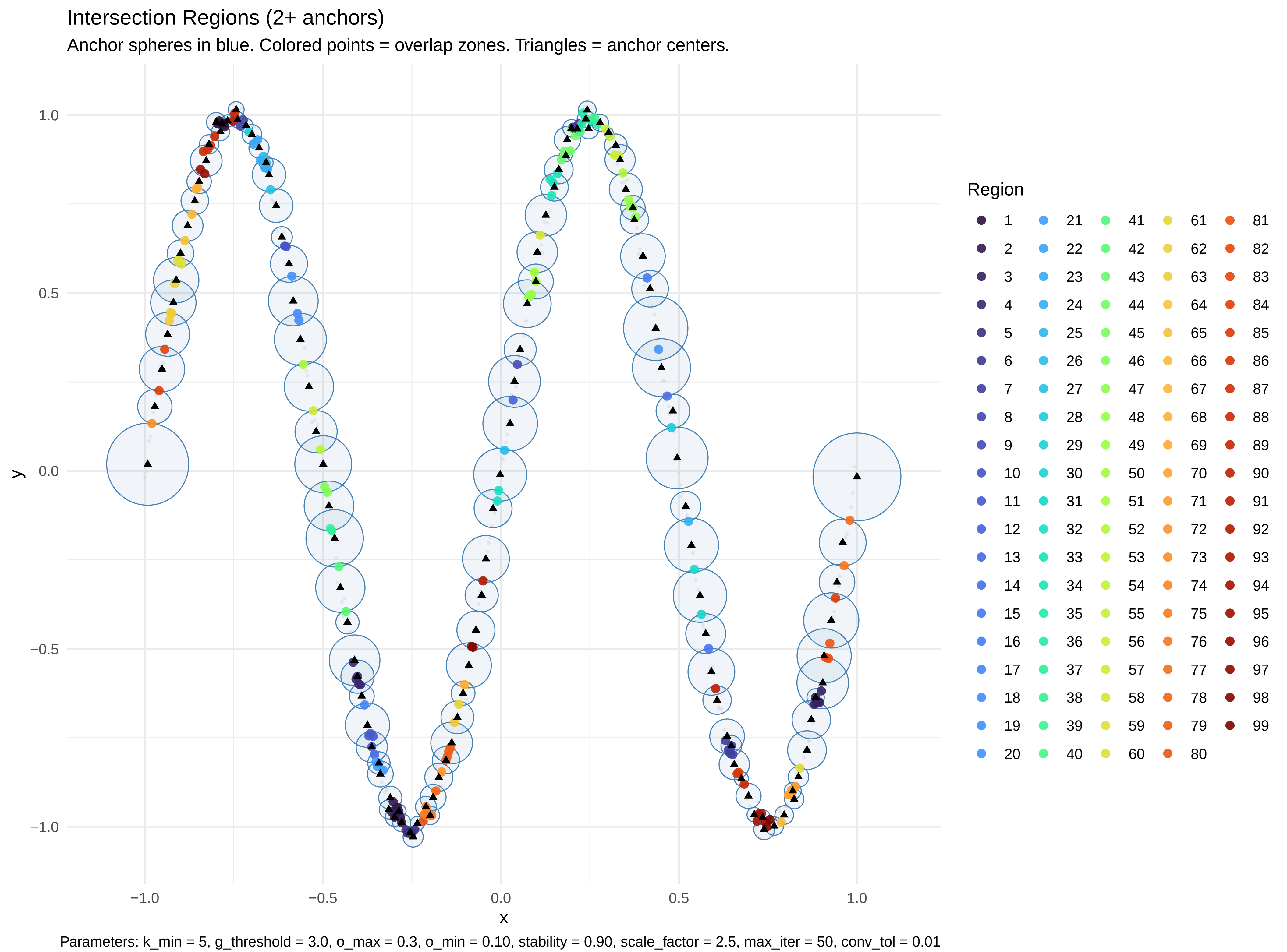
Point Density Distribution
1.0
0.5
0.0
−0.5
−1.0
y
−1.0
−0.5
0.0
0.5
1.0
x
density
200
100

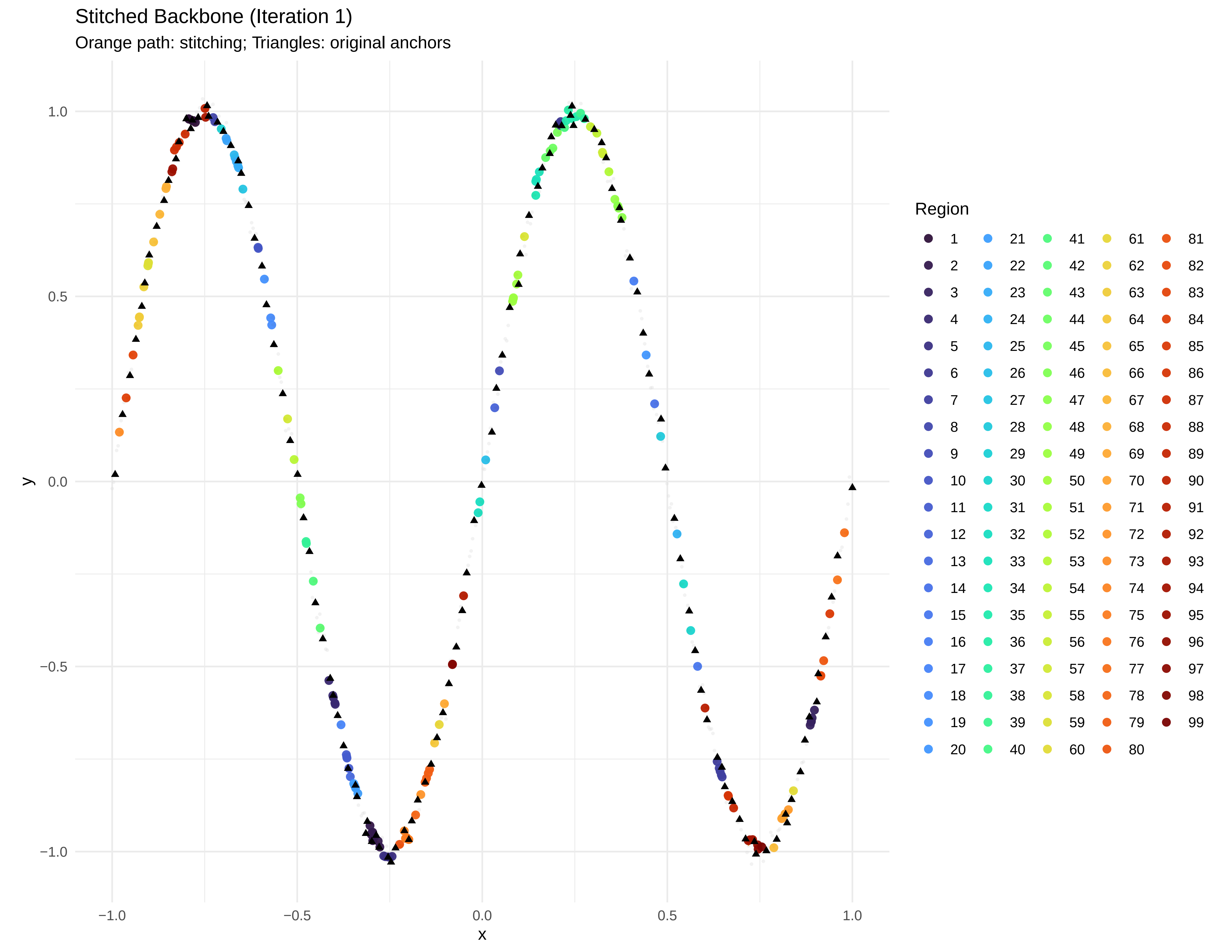

Global Tangent Field (PC1)
Density
200
100
1.0
0.5
0.0
−0.5
−1.0
y
−1.0
−0.5
0.0
0.5
1.0
x



Atlas Selection (Anchors)
1.0
0.5
0.0
−0.5
−1.0
y
−1.0
−0.5
0.0
0.5
1.0
x
density
200
100


Anchor SVD Segments
Same anchors as plot 4: spheres first, then their SVD (PC1) in red
1.0
0.5
0.0
−0.5
−1.0
y
−1.0
−0.5
0.0
0.5
1.0
x


Intersection Regions (2+ anchors)
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.
y
1.0
0.5
0.0
−0.5
−1.0
x
−1.0
−0.5
0.0
0.5
1.0
Region
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01


Stitched Backbone (Iteration 1)
Orange path: stitching; Triangles: original anchors
y
x
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Region
1
2
3
4
5
6
7
8
9
10
11
12
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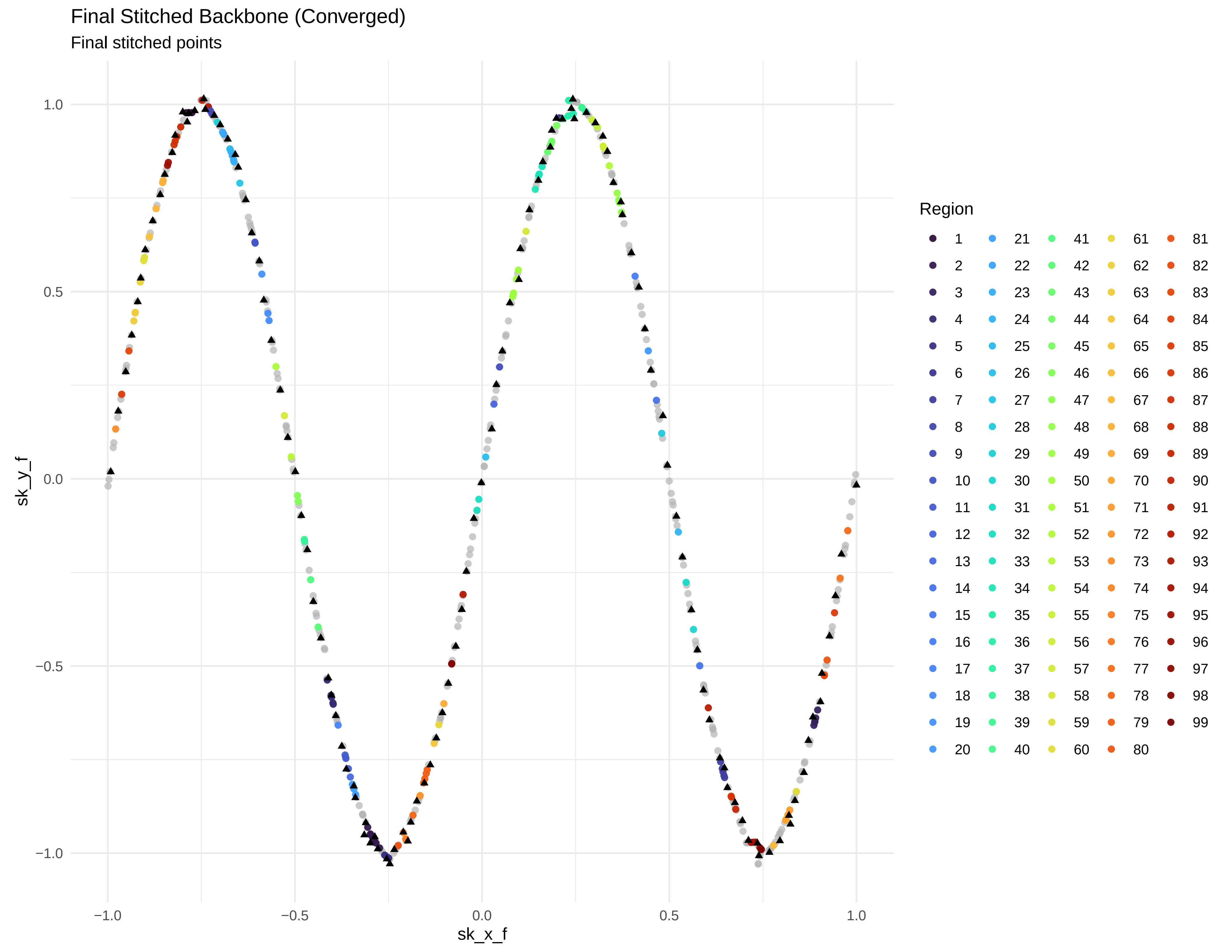
98
99


Final Stitched Backbone (Converged)
Final stitched points
sk_y_f
sk_x_f
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Region
1
2
3
4
5
6
7
8
9
10
11
12
13
14
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16
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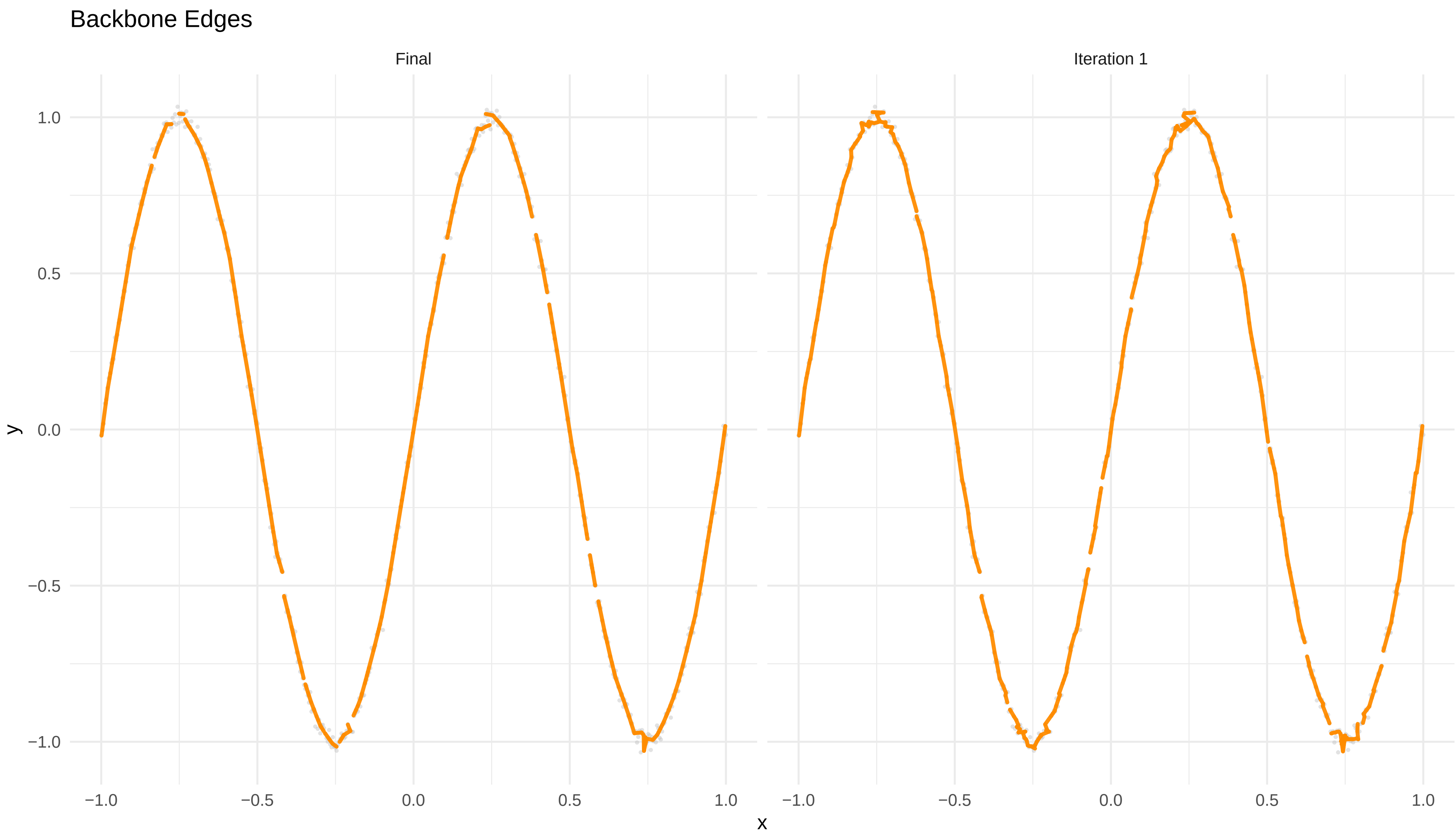
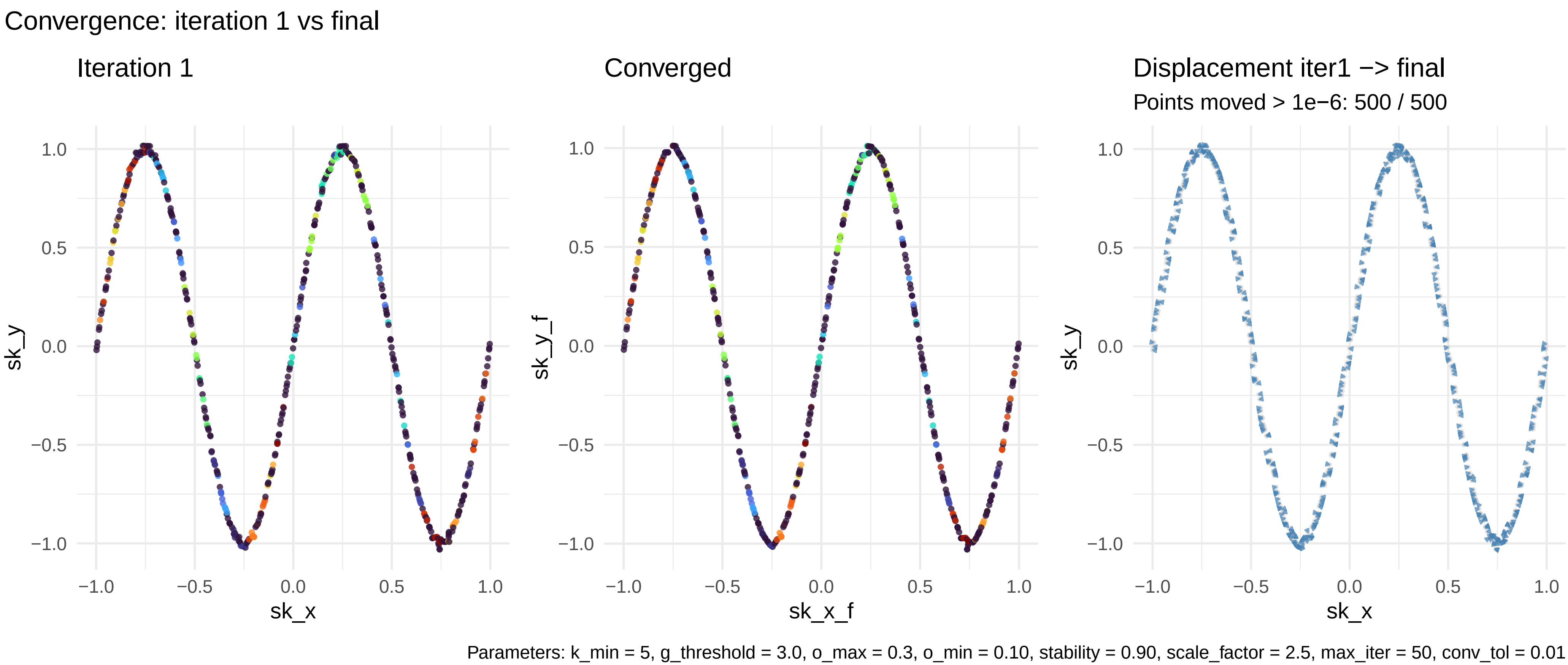
97
98
99


Backbone Edges
Final
Iteration 1
1.0
0.5
0.0
−0.5
−1.0
y
−1.0
−0.5
0.0
0.5
1.0
x



Convergence: iteration 1 vs final
Iteration 1
Converged
Displacement iter1 −> final
Points moved > 1e−6: 500 / 500
sk_y
sk_x
sk_y_f
sk_x_f
1.0
0.5
0.0
−0.5
−1.0
Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01